FACULTY OF SCIENCES

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Lab 10 - Solving an old exam

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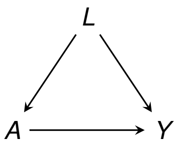
Question 1 - Setup

- Average causal effect of A on Y.
- We have covariates L sufficient to adjust for confounding (and possibly more).
- Use random forest regression to predict Y based on L, separately for $A = 1$ and $A = 0$ (one model for the treated, one for the controls).
- Average the difference of the predictions (this is G-computation):

$$\frac{1}{n}\sum_{i=1}^{n} \hat{E}^{rf}[Y \mid A_i = 1, L_i] - \hat{E}^{rf}[Y \mid A_i = 0, L_i]$$

Question 1 (a) - Appropriate Selection of Confounders



- When fitting the two outcome models, the selected confounders are those that help predicting the outcome Y.
- However, there may be some confounders that are strongly associated with the exposure A and only a small association with the outcome Y.
- In this last case, those confounders may be excluded from the outcome model, thus leading to an incorrect selection of confounders.

Question 1 (a) - Appropriate Selection of Confounders

		Y		
		Zero	Small	Strong
A	Zero	Irrelevant	Irrelevant	Irrelevant
	Small	Irrelevant	Greyzone	Important
	Strong	Irrelevant	Important	Essential

Table: Appropriate selection of confounders based on the relationship between L and A, Y. (Table taken from Michael Knaus).

- Relying only on the outcome model, we may miss those confounders having a small association with Y but strong with A.

Question 1 (b) - Bootstrap

- The bootstrap replaces the unknown P with the empirical P_n. Since $P_n - P = O_p(n^{-1/2})$, P_n is only a small perturbation of P.
- The bootstrap standard error can be computed as:

$$\widehat{\mathrm{SE}}(\hat{\psi}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\psi}^{(b)} - \bar{\psi}\right)^2}$$

and generally correctly approximates $\mathrm{Var}_P(\hat{\psi})$, yielding a valid standard error.

Question 1 (b) - Why the Bootstrap Fails Here

The g-computation estimator with random forests is

$$\hat{\psi} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}(1, L_i) - \hat{\mu}(0, L_i)\right].$$

- Random forests are highly adaptive: even small perturbations of the data can lead to substantial changes in $\hat{\mu}$.
- Consequently, the variability of $\hat{\psi}^{(b)}$ across bootstrap samples (from P_n) does not (always) reliably approximate $\mathrm{Var}_P(\hat{\psi})$.
- The bootstrap SE measures variability under P_n, but we need variability under P, which these coincide only for sufficiently regular estimators.

Question 1 (c) - Debiased Estimators

Since the bias affecting a plug-in estimator of a model-free estimand is primarily driven by (-) the sample average of the estimand's efficient influence curve...

- Derive the efficient influence function of the estimand (i.e, the ATE).
- Use a debiased estimator that sets its sample average to zero.
- Valid inference can be obtained by $(1/\sqrt{N})^* \, \mathrm{sd}(\mathrm{EIC})$.

Question 2 (a) - Setup

We consider Assumption-Lean Logistic Regression:

$$\text{logit}\left\{E\left(Y^a \mid L\right)\right\} = \alpha(L) + \beta a$$

- Y: dichotomous outcome.
- A: dichotomous exposure (randomly assigned).
- L: baseline characteristics.

Question 2 (a) - Assumption-Lean Logistic Regression

For a dichotomous exposure A, we consider weighted averages

$$\frac{E\{w(L)\beta(L)\}}{E\{w(L)\}}$$

with $\beta(L) = g\{E(Y \mid A = 1, L)\} - g\{E(Y \mid A = 0, L)\}$.

We know that the treatment is randomized, thus $w(L)$ is constant.

The estimand reduces to

$$\frac{E\{w(L)\beta(L)\}}{E\{w(L)\}} = E[\beta(L)]$$

Question 2 (a) - Assumption-Lean Logistic Regression

Note that $\beta(L)$ is defined as:

$$\begin{aligned}\beta(L) &= \text{logit}(E[Y^1 \mid L]) - \text{logit}(E[Y^0 \mid L]) \\ &= \log\left(\text{odds}(Y=1 \mid A=1, L)\right) - \log\left(\text{odds}(Y=1 \mid A=0, L)\right) \\ &= \log\left(\frac{\text{odds}(Y=1 \mid A=1, L)}{\text{odds}(Y=1 \mid A=0, L)}\right)\end{aligned}$$

Recall that for a probability p, $\text{odds}(p) = \frac{p}{1-p}$, so $\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$.

Thus, in the randomized setting (constant weights), the assumption-lean logistic regression targets the average conditional log-odds ratio, i.e., $E[\beta(L)]$.

- If $\beta(L)$ is constant, homogeneous treatment effect on the log odds scale.
- If $\beta(L)$ is not constant, using assumption-lean logistic regression we are still targeting the mean of the conditional log-odds ratio.

Question 2 (b) - Non Collapsibility

Is the average of the conditional log-odds ratio equal to the marginal log-odds ratio?

$$E\left[\log\left(\frac{\text{odds}(Y=1 \mid A=1, L)}{\text{odds}(Y=1 \mid A=0, L)}\right)\right] \stackrel{?}{=} \log\left(\frac{\text{odds}(Y=1 \mid A=1)}{\text{odds}(Y=1 \mid A=0)}\right)$$

- Non-collapsibility is a property of some statistical measures of association that causes them to change when adjusting for a covariate, even if that covariate is not a confounder.
- Other measures are collapsible: for instance, risk differences.

Question 2 (b) - Non Collapsibility

- Assume that $\beta(L) = \beta$, i.e. the treatment effect is constant. This means that all the conditional log-odds ratio are the same.
- However, the marginal log-odds ratio can still be different due to non-collapsibility: conditioning on covariates changes the magnitude.
- This happens due to the ‘mathematical’ way the (log-)odds ratio are defined.

QUESTION 2 (B) - NON COLLAPSIBILITY

Conditional and marginal odds ratios

	Counterfactual risk (placebo)	**Counterfactual risk (treatment)**	**Causal odds ratio**
Men (25%)	0.5	0.75	3.0
Women (75%)	0.25	0.5	3.0
Overall	0.3125	0.5625	2.83

Table: Huitfeldt A, Stensrud MJ, Suzuki E. On the collapsibility of measures of effect in the counterfactual causal framework. Emerg Themes Epidemiol. 2019 Jan 7;16:1. doi: 10.1186/s12982-018-0083-9. PMID: 30627207; PMCID: PMC6322247.

Noncollapsibility of the OR is the phenomenon whereby the crude OR from the marginal table cannot be expressed as the weighted average of the stratum-specific ORs even in the absence of confounding.

Question 2 (b) - Non Collapsibility

If instead we look at risk differences:

$$\beta(L) = E(Y \mid A = 1, L) - E(Y \mid A = 0, L)$$

$$\begin{aligned} E[\beta(L)] &= E\big[E(Y \mid A = 1, L) - E(Y \mid A = 0, L)\big] \\ &= \underbrace{E[Y \mid A = 1] - E[Y \mid A = 0]}_{\text{marginal RD}} \end{aligned}$$

- Unlike the odds ratio, the risk difference is collapsible.
- Averaging the conditional risk differences over the population recovers the marginal risk difference.

Question 2 (c) - Estimating the propensity score

- $0.5 \longrightarrow$ Yes, since we know it by design.
- $\sum_{i=1}^{n} A_i / n \longrightarrow$ Yes, because we know it is constant (in an observational study we could not estimate the propensity score with a sample average).
- $\hat{E}^{rf}(A \mid L) \longrightarrow$ Yes, since in the general case we would estimate it using data-adaptive estimators.

Question 3 (a) - Orthogonal Learners

Disadvantages of T-Learner

- It does not target the CATE, but instead consists of two separate minimization problems resulting in two, possibly regularized, outcome predictions.
- Risk of extrapolation if the distribution of the covariates is different for treated and controls.
- It is not orthogonal (see next slide to understand).

T-Learner

The prediction problems do not know of joint goal to approximate a difference

$$\Rightarrow \hat{\mu}_1(X) \text{ minimizes } \mathrm{MSE}(\hat{\mu}_1(X)) = \mathbb{E}\left[(\hat{\mu}_1(X) - \mu_1(X))^2\right]$$
$$\Rightarrow \hat{\mu}_0(X) \text{ minimizes } \mathrm{MSE}(\hat{\mu}_0(X)) = \mathbb{E}\left[(\hat{\mu}_0(X) - \mu_0(X))^2\right]$$

BUT what they should aim to minimize is

$$\begin{aligned}
\mathrm{MSE}(\hat{\tau}(X)) =& \mathbb{E}\left[(\hat{\tau}(X) - \tau(X))^2\right] \\
=& \mathbb{E}\left[(\hat{\mu}_1(X) - \hat{\mu}_0(X) - (\mu_1(X) - \mu_0(X)))^2\right] \\
=& \mathbb{E}\left[(\hat{\mu}_1(X) - \mu_1(X))^2\right] + \mathbb{E}\left[(\hat{\mu}_0(X) - \mu_0(X))^2\right] \\
& -2\mathbb{E}[(\hat{\mu}_1(X) - \mu_1(X))(\hat{\mu}_0(X) - \mu_0(X))] \\
\\
=& \mathrm{MSE}(\hat{\mu}_1(X)) + \mathrm{MSE}(\hat{\mu}_0(X)) - 2\,\mathrm{MCE}(\hat{\mu}_1(X), \hat{\mu}_0(X))
\end{aligned}$$

where MCE is Mean Correlated Error.

Question 3 (a) - Orthogonal Learners

Advantages of Orthogonal Learners (DR-Learner, R-Learner,...)

- They directly target (a weighted) CATE by minimizing the counterfactual prediction error:

$$E\left[\lambda\left\{\pi(L)\right\}\left\{Y^1 - Y^0 - \psi(L)\right\}^2\right]$$

- They are orthogonal, meaning that even if you use data-adaptive estimator for nuisance parameters, you have asymptotically the same properties as if they were given.
- DR-Learner is orthogonal because it relies on the efficient influence function of the ATE.
- R-Learner achieves orthogonality by residualization.

Question 3 (b) - Orthogonal Learners in Randomized Experiments

In both the orthogonal learners, there are 2 nuisance parameters:

- Outcome model
- Propensity score

If we are in a randomized setting, there is no need to estimate the propensity score, since it is constant and known.

This is especially useful in the context of DR-Learner, since it relies on the positivity assumption of the propensity score.

Question 3 (c) - (Naive) MSE among the Treated

We are given $\psi(L)$ for $E\left(Y^1 \mid L\right)$. We would like to estimate

$$E\left[\left\{Y^1 - \psi(L)\right\}^2\right]$$

as the sample average of $\{Y - \psi(L)\}^2$ among treated individuals.

Is it valid?

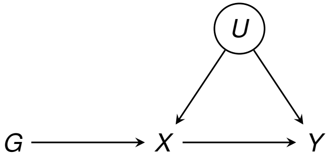
- In an observational study, the characteristics of the treated and controls are different. Thus estimating this quantity only on the treated subsample would not give a valid estimate of the marginal mean squared error.
- However, in a randomized setting, it would be valid (note that in this case also the ATE = ATT).

Question 3 (d) - (Debiased) MSE

- Let $Z^1 = \left(Y^1 - \psi(L)\right)^2$
- We are interested in estimating $E\left[Z^1\right]$
- The efficient influence function is

$$\frac{A}{P(A=1 \mid L)}\left\{Z - E[Z \mid A=1, L]\right\} + E[Z \mid A=1, L] - E\left[Z^1\right]$$

Question 4 (a) - Causal Diagram



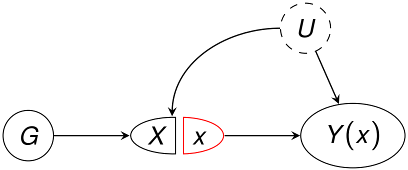
where

- Y = Alzheimer's disease (outcome),
- X = Diabetes (exposure),
- G = HLA-DR3 genotype (IV),
- U = ... (unobserved confounders)

Question 4 (b) - Test in presence of unmeasured confounders

- We want to test whether diabetes affects the risk of Alzheimer's disease.
- However, since we have unmeasured confounders, we can not carry out a naive analysis between those two variables.
- We know that the effect of the IV is *only through* the exposure.
- We can run a simple linear regression of $Y \sim G$ and thest whether its coefficient is different from 0.

Question 4 (c) - SWIG and Independence Assumption



The independence assumption:

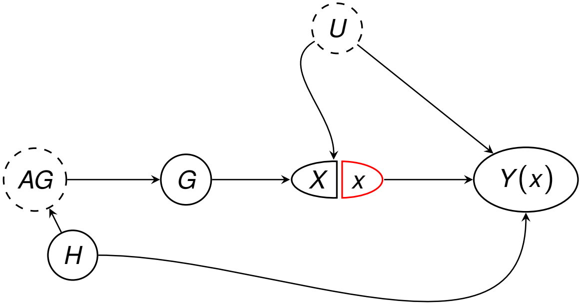
$$Y^x \perp\!\!\!\perp G$$

says that there is no direct effect of the instrument to the outcome and there are no unobserved confounders of G and Y (randomization).

Question 4 (c) - SWIG and Independence Assumption

- Is that assumption testable?
- No, since Y^x is a counterfactual, thus independence is not empirically testable. It is an assumption required for identification using G as an IV.

Question 4 (d) - Covariates H



- $Y(x)$ = counterfactual Alzheimer's disease under $X = x$
- X = Diabetes (exposure, intervened upon)
- G = HLA-DR3 genotype (IV)
- U = unmeasured confounders
- H = family history of Alzheimer's disease
- AG = ancestral genotypes

Question 4 (d) - Covariates H

- H is a confounder of G and Y, thus not controlling for it violates the independence assumption.
- We will revisit the assumption as conditional independence:

$$Y^x \perp\!\!\!\perp G \mid H$$

QUESTION 4 (D) - IV ESTIMATION

- Consider the model

$$E(Y \mid X, H, G, U) = \psi X + g(H, U)$$

- We can define ψ by the estimand:

$$\psi = \frac{E\{\mathrm{Cov}(G, Y \mid H)\}}{E\{\mathrm{Cov}(G, X \mid H)\}}$$

If instead the model is misspecified (ψ is not constant):

$$E(Y \mid X, H, G, U) = \psi(H)X + g(H, U)$$

our estimand reduces to

$$\psi = \frac{E\{\mathrm{Cov}(G, Y \mid H)\}}{E\{\mathrm{Cov}(G, X \mid H)\}} = \frac{E\{w(H)\psi(H)\}}{E\{w(H)\}}$$

for weights $w(H) = \mathrm{Cov}(G, X \mid H)$.

Question 4 (d) - IV Estimation

We can then estimate ψ using debiased machine learning:

- **Efficient influence curve**

$$\frac{\{G - E(G \mid H)\}[Y - E(Y \mid H) - \psi\{X - E(X \mid H)\}]}{E[\{G - E(G \mid H)\}\{X - E(X \mid H)\}]}$$

- **A debiased estimator that sets its sample average to zero**

$$\hat{\psi} = \frac{\sum_{i=1}^{n} \left\{G_i - \hat{E}\left(G_i \mid H_i\right)\right\} \left\{Y_i - \hat{E}\left(Y_i \mid H_i\right)\right\}}{\sum_{i=1}^{n} \left\{G_i - \hat{E}\left(G_i \mid H_i\right)\right\} \left\{X_i - \hat{E}\left(X_i \mid H_i\right)\right\}}$$

- **Valid inference** can be obtain by the usual $\frac{1}{\sqrt{N}} * \text{sd}(\text{EIC})$

Question 4 (d) - IV Test

A better way to test whether diabetes affects the risk of Alzheimer's disease is obtained by testing:

$$\theta = E\{\mathrm{Cov}(G, Y \mid H)\}$$

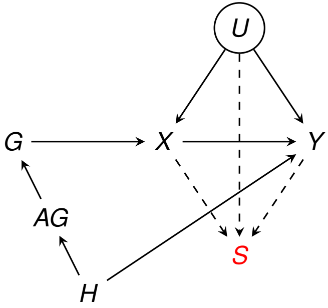
The efficient influence curve of θ*is*

$$\{G - E(G \mid H)\}\{Y - E(Y \mid H)\} - \theta.$$

Which leads to

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} \left\{G_i - \hat{E}\left(G_i \mid H_i\right)\right\} \left\{Y_i - \hat{E}\left(Y_i \mid H_i\right)\right\}$$

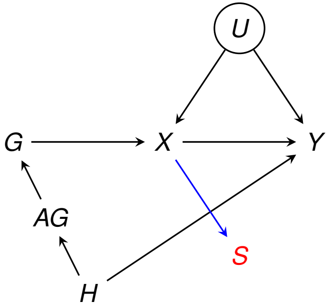
Question 4 (e) - Selection Bias



where

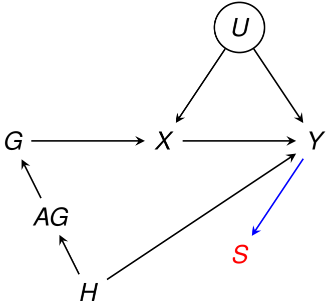
- $S =$ Binary Live Status
- Other variables as before

QUESTION 4 (E) - SELECTION BIAS (I)



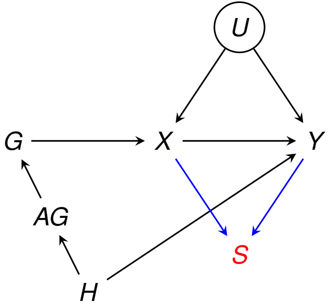
- S is a child of X (which is a collider of G and U).
- Conditioning on S would be similar to conditioning on X, thus creating a spurious association between G and U.
- This would cause bias since there will be a path from G to Y through the unobserved factors U, leading to a violation of the exclusion restriction (regardless of whether X affects Y).

Question 4 (e) - Selection Bias (ii)



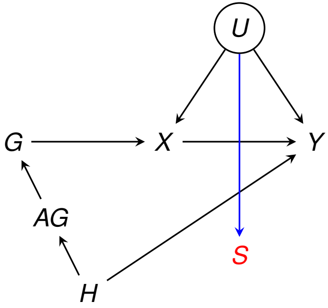
- This is another form of selection bias called case-control bias. (Model 18 in Cinelli, Carlos, Andrew Forney, and Judea Pearl. "A crash course in good and bad controls." Sociological Methods & Research 53.3 (2024): 1071-1104.) (Figure 11.12 in Chernozhukov, Victor, et al. "Applied causal inference powered by ML and AI." arXiv preprint arXiv:2403.02467 (2024).)
- It breaks the randomization assumption of the instrument $G \not\perp\!\!\!\perp Y^X$

Question 4 (e) - Selection Bias (iii)



- Conditioning on S induce collider bias.
- If there was no relationship between X and Y, conditioning on S may induce it.
- If there is an effect from X to Y, conditioning on S may introduce bias in the estimation.

Question 4 (e) - Selection Bias (iv)



- Since S is not affected by other variables, conditioning on it is not problematic for the analysis.